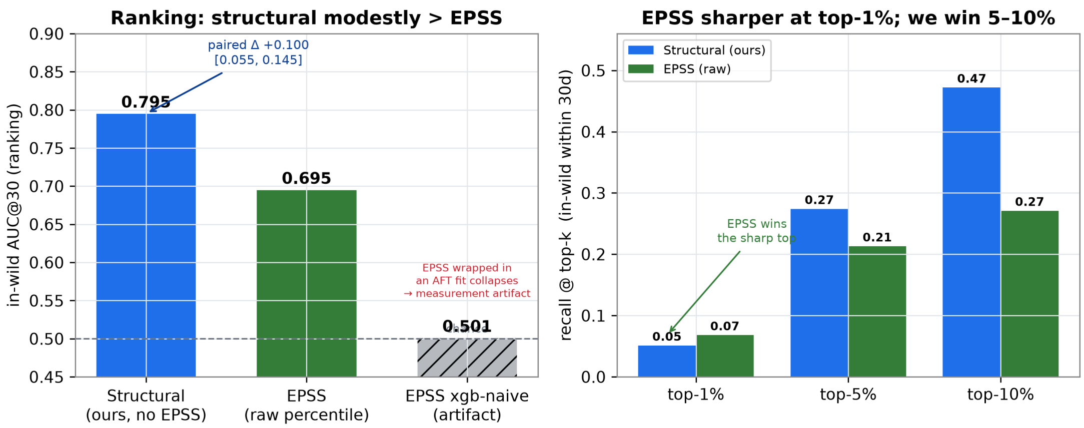
Temporal Exploit Prediction — Progress Update

Predicting *when* a published CVE becomes weaponized (survival analysis; complements EPSS) · week of 23–30 June 2026 · supervisor meeting, 1 July 2026

Headline. The in-wild head was re-targeted into a *measurable* comparison against EPSS and adversarially verified: at disclosure (cold-start), structured publication-time features out-rank EPSS by **+0.100 AUC@30** (95% CI [0.055, 0.145], wins 13/15 time-splits); PR-AUC is tied. The project is now consolidated into a 19-page, fully-cited report.

Same target as EPSS: a modest ranking win + a precision tie — not a blowout



*In-wild ranking on the same target as EPSS (1,310 test events, 15 walk-forward origins). **Left:** structural features (0.795) modestly beat raw-EPSS (0.695); wrapping EPSS in a model collapses it to chance (0.501) — the baseline bug we fixed. **Right:** we win at top-5/10% depth, EPSS is sharper at the very top-1%. Source: artifacts/inwild_epss_parity.json.*

What's new since last week

- **Concrete win over EPSS at t=0:** structural AUC@30 **0.795** vs raw-EPSS **0.695** (Δ+0.100); AUC@90 Δ**+0.134** (15/15); top-decile recall **0.47** vs 0.27. Value = the cold-start regime where EPSS is near-chance / ~48% missing.
- **GreyNoise telemetry connector built** (verified API, TDD, 6 tests, passed an RE audit) — next lever for scarce in-wild labels. Caveat: *prospective-only* (rolling ≤30d, no backfill).
- **Report is literature-grounded:** 40 verified citations + References integrated into the 19-page PDF (checked against DOI / arXiv / venue).

Why the result is trustworthy (rigor corrections)

- **Fixed a baseline bug that would have inflated the win:** EPSS must be ranked by its raw score (in-model = 0.501, chance). The +0.100 is measured against a *strong* EPSS.
- **Corrected the target framing (honesty):** the "in-wild" label is ~93% VulnCheck catalog-add dates — an administrative proxy (median ~+175d lag; ~22% within 30d), not true onset. Claim = "cold-start ranking advantage," not "predicts real-world exploitation." Adversarial RE (2 agents): no leakage / handicap.
- **Caught 3 errors in the report's own draft citations** (wrong year, samples-vs-events unit, wrong journal volume); flagged 1 non-peer-reviewed source.

Decisions to discuss

- **Framing:** is "cold-start ranking advantage over EPSS" the right thesis claim, given the proxy label (+175d lag)?
- **Telemetry:** GreyNoise is forward-only — wire it for prospective evaluation, or accept the retrospective ~396-event ceiling as the wall?
- **Dissertation angle:** continue toward the epidemiological mixture model for pre-disclosure dynamics?
- **Logistics:** merge the `inwild-epss-parity` branch to master.

Status: 337 tests green · report 19pp / 40 refs · parity work pushed (branch `inwild-epss-parity`)	**Sources:** artifacts/inwild_epss_parity.json; commits 79d4213, cf3348d, 872b588 (parity), bace8e8/fdb2541 (GreyNoise); docs/project_report_2026-06-23.pdf